

Marco Sabatini

Professional Summary

Engineering leader with 15+ years of experience in building scalable data platforms, distributed systems, and cloud infrastructure across fintech, cybersecurity, and travel-tech industries. Proven track record in leading engineering teams, designing modern data architectures, and delivering production-grade data platforms for analytics, risk modeling, and data-driven decision making. Strong expertise in data pipelines, big data ecosystems, cloud-native infrastructure, and platform engineering. Passionate about mentoring engineers, building high-performing teams, and aligning technical platforms with business strategy.

Core Competencies

Data Platform Architecture

- Data Lakehouse architectures
- Data ingestion and processing pipelines
- Streaming and event-driven systems
- Data lifecycle management
- Data quality and reliability

Cloud Data Platforms

- AWS & Azure storage solutions
- AWS & Azure RDS databases
- Kubernetes container orchestration
- Infrastructure as Code with Terraform

Data Engineering & Technologies

- Apache Spark for big data processing
- Databricks for data analytics
- SQL and NoSQL databases (Oracle, MySQL, PostgreSQL)
- Elasticsearch and SOLR for search
- Data orchestration and microservices architecture
- Programming: Python, Java, Ruby, TypeScript, Bash

Leadership & Strategy

- Engineering team leadership and management
 - Platform strategy and architecture design
 - Cross-team technical coordination
 - Stakeholder alignment with product and business teams
 - Mentoring and developing engineering talent
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Professional Experience

Associate Director – Software Engineering

UBS – Lugano, Switzerland

November 2024 – Present

Lead a team of 15 engineers responsible for migrating financial risk models to cloud-native data platforms supporting enterprise analytics and risk management.

Key Achievements

- Led migration of US mortgages probability of default and loss given default risk modeling workloads from on-premises to Azure cloud platforms, managing a portfolio of 100,000 mortgages across the entire US
- Designed scalable architecture using Azure Kubernetes, Databricks, and cloud storage
- Implemented Infrastructure as Code with Terraform for standardized deployments
- Improved scalability and reliability of data processing pipelines for quantitative team of 5 people
- Coached four engineers and promoted engineering best practices across the team

Technologies: Azure Storage, Kubernetes, Databricks, Python, Terraform, Docker, Oracle Database, Git-Lab

Senior Software Architect

Denexus – Lugano, Switzerland

May 2022 – November 2024

Architected and developed cyber risk intelligence platform for aggregating and enriching large-scale network asset and vulnerability data.

Key Achievements

- Designed cloud-native platform for cybersecurity data ingestion, enrichment, and indexing
- Built scalable data pipelines using Python and Databricks for intelligence data processing
- Implemented microservices in Python using Fast API for data discovery and search across industrial networks
- The platform calculates risk for approximately 20 sensitive industrial plants in the USA and Europe, involving ingestion of terabytes of data and optimized model execution in minutes
- Collaborated with leadership on platform architecture and roadmap development

Technologies: AWS (ECS, EC2, S3), Databricks, Python, FastAPI, Terraform, Docker, PostgreSQL, Splunk

Senior Software Engineer

Nozomi Networks – Mendrisio, Switzerland

November 2020 – May 2022

Developed data services for industrial cybersecurity analytics platforms.

Key Achievements

- Built systems for collecting, processing, and enriching vulnerability intelligence data
- Led security research engineering team for vulnerability workflows
- Collaborated with SRE teams on AWS infrastructure using Terraform
- Improved data reliability and operational monitoring

Technologies: AWS, Terraform, Elasticsearch, SOLR, Docker, Ruby, JavaScript, MySQL

Tech Lead

lastminute.com – Chiasso, Switzerland

November 2018 – November 2020

Technical leader in travel platform engineering organization.

Key Achievements

- Led backend teams developing high-scale travel microservices
- Coordinated cross-team optimization using production KPIs
- Supported migration to cloud-based infrastructure
- Mentored engineers and promoted best practices

Technologies: Kotlin, Java, MySQL

Senior Agile Software Engineer

lastminute.com – Chiasso, Switzerland

November 2015 – November 2018

Engineer on Big Data pricing analytics platform team.

Key Achievements

- Designed and developed data pipelines for flight pricing analytics
- Built distributed processing workflows using Spark on Google Cloud
- Contributed to migration from monolithic to microservices architecture
- Supported data-driven pricing optimization with product teams

Technologies: Scala, Apache Spark, Google BigQuery, Dataproc, Kotlin, Java

Earlier Experience

Solution Architect Consultant

Wedjaa – Rome, Italy

2011 – 2015

Worked as a consultant on multiple enterprise projects, leading client engagement, technical assessment, solution architecture, and software delivery. Responsibilities included defining architectures, developing software, and forming or shaping internal and external development teams.

Major projects during this period included: - A Java-based ticketing system for an Italian cinema chain with approximately 30 locations and hundreds of screens, supporting peak holiday loads of 20,000 daily bookings. - An online medical reporting platform for the Lazio region, integrating Alfresco and the open-source eHealth Integration Framework (IPF) for patient data management. - A project reporting and accounting platform for an international telecom company, managing requirements definition and Java development with Jira integration.

System Analyst

IDS – Rome, Italy

2009 – 2011

Contributed to the development of two aerospace systems for flight procedure design and obstacle management in controlled aerospace. Responsibilities included requirements gathering, feature development, and production releases. Developed most of the application backend in Java, with frontend components in JavaScript and HTML. The software was deployed to production and used by approximately ten national and international aviation and airport authority organizations.

Software Developer

IBM – Rome, Italy

2008 – 2009

Developed a Java-based search engine using *Apache Lucene* for information retrieval. As a junior software engineer working alongside a senior engineer, I helped define and implement features to replace a proprietary search engine for a publishing lifecycle system serving international magazines. The solution indexed millions of documents per client in both batch and real-time modes, using Java 1.4 version with socket-based integration to the publishing platform. The product went into production for around thirty customers.

Education

Master of Science in Computer Science

University of Rome Tor Vergata, Italy

Thesis: Integration of Legacy Systems Using SOA Principles

Grade: 107/110

Professional Development

- Advanced Test Driven Development – J.B. Rainsberger
- Surviving Legacy Code – J.B. Rainsberger
- Applied Functional Programming – Avanscoperta
- Machine Learning & Predictive Analytics – Simone Scardapane
- Secure Coding – Nozomi Networks